DOCKHUB · DESIGN PRECIS

DockHub design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: R1 to R16 are PR-001 to PR-016 in DKH-PRS-001 (Was column); concept options DKH-OPT-001 (option A recommended, awaiting Amish); fan rule on aerosol discharge (DKH-DEC-001 D1); the design itself is unchanged
0.8	2026-10-02	Amish Chadha	Approved follow-ups carried into the design: anchors and site pad (BOM 17, 18; R15 met), vermiculite floor trial option (BOM 19), anonymous prepaid cards in the token list, flattest-pack solar sensitivity; numbers from DKH-CAL-001 v0.6
0.7	2026-10-02	Amish Chadha	R6, R9, pack ownership, pilot partner, propagation test, suppression, siting and anonymous cards as decided on 2026-10-02 (DKH-DEC-001)
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-10-01	Amish Chadha	Constructable design (DKH-DDR-003, Draft) and prototype build plan DKH-

			BLD-001; components, mass, cost and R16 status updated
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. Record the TRL 2 review items adopted under Amish's 2026-09-25 instruction (DKH-DDR-001); SwapCell interface v0.3; numbers checked against DKH-CAL-001; media from the parametric model
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

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Kit 1.9.1 note, 2026-10-31. The requirement numbers R1 to R16 below are now PR-001 to PR-016 in DKH-PRS-001 (its Was column maps them; the calculation note and build plan still use R numbers). This precis describes option A, swap bays in a steel cabinet; the concept options are scored in DKH-OPT-001 and the choice is proposed, awaiting Amish (DKH-DDR-004). One rule is added by the STEM check (DKH-STM-001 S13, DKH-DEC-001 D1): on an aerosol discharge the roof fans stop and the plenum vents through the passive hood slot.

Summary

DockHub is a sidewalk cabinet, 1.0 m wide and 0.5 m deep, with four lockable steel bays that each hold and charge one SwapCell pack. A rider taps a card or phone, returns a flat pack to the empty bay and takes a charged one from another, in about 31 seconds. Each bay is a copy of the SwapCell wall dock built to SwapCell interface v0.3: a cradle with the blind-mate receptacle and its 10 kΩ INTERLOCK coding resistor, a certified 54.6 V 5 A charger and a CAN channel that checks the pack before charging. Every bay is its own steel compartment venting to a rear plenum, so a pack fault is detected, cut off and ducted away from the user side. A 400 W solar canopy shades the front and feeds bay 1; the grid supplies the rest through one single-phase circuit.

The sizing note DKH-CAL-001 confirms the swap time, charge speed, throughput (with four bays), grid draw, reach heights and footprint. It also showed two requirements not met: the canopy supplies 9.3 % of the charging energy against the relaxed 10 % target (R6), and the pack's 45 °C charge limit stops charging whenever ambient air is above about 29 to 39 °C (R9). On 2026-10-02 Amish restated R6 as about 9 % and R9 as full-rate charging up to about 30 °C ambient, with daytime pauses accepted (DKH-DEC-001); R6 is now met on paper and R9 at risk. Making the design buildable (DKH-DDR-003) took the estimated cost to $1,214, and the anchors and cast site pad chosen for the first pilot on private ground add $305, so the two-bay prototype is estimated at $1,519 against the $1,200 value-engineering target (R16, $319 over). With the anchors and pad, anchoring (R15) is met on paper; fire containment (R8) remains at risk. Because a swap needs an empty bay, the four-bay station runs with three packs and the two-bay first prototype with one (DKH-REQ-001, DKH-DDR-001).



Figure 1. DockHub on a sidewalk with a 1.75 m person for scale, four-bay fit with three packs and one empty bay. Rendered from the parametric model `cad/src/model.py` *; concept, not for fabrication.*

How it works

1. **Identify.** The rider taps an NFC card or phone on the access panel (BOM 7). The controller (6) checks the token against its local list, so the station works without a network connection. The list holds named tokens and anonymous prepaid cards sold by the partner, each card carrying a pack deposit and a number of prepaid swaps; an anonymous card is known to the station only by a hash of its identifier.
2. **Return.** The empty bay's door (2) unlocks. The rider places the flat pack connector down in the cradle (3) and closes the door. The receptacle mates ground first, then power and CAN, and the interlock last. The 10 kΩ coding resistor in the INTERLOCK loop wakes the pack, as SwapCell interface v0.3 requires (item W).
3. **Check.** The controller sends the SwapCell heartbeat as a dock (host type 1) on that bay's CAN channel. The pack reports its identity, limits, temperatures and state of health. A pack with a fault, a damage flag or a state of health below 70 % stays locked in and is flagged for a technician; the rider still gets a charged pack.
4. **Release.** The door of the fullest healthy pack at 80 % state of charge or more unlocks; the rider lifts it out by its handle and closes the door. That bay becomes the empty bay for the next rider. The controller logs the pair of pack IDs against the rider token.

5. **Charge.** When the pack's limits allow, the controller closes that bay's DC contactor and its charger (5) runs CC-CV to 54.6 V. Bay 1 charges from the solar panel (11) through the MPPT controller (13) whenever the panel offers 50 W or more, and from its AC charger otherwise.
6. **Watch.** A thermistor at every receptacle, the pack's own temperature reports and a heat and smoke detector (9) are read every second. On an alarm the bay contactor opens, fans (10) run, the bay door stays locked, a light and buzzer warn people to stand back and an alert goes out by LTE-M.

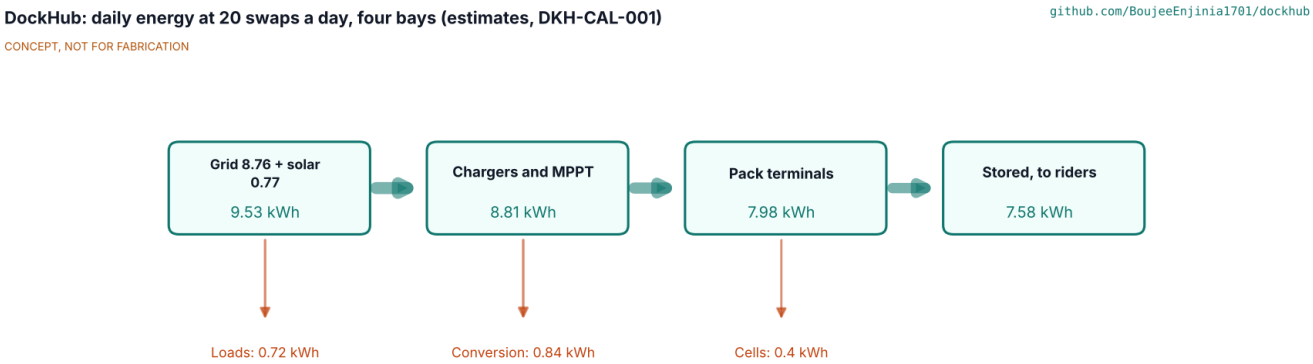


Figure 2. Daily energy at the design duty case of 20 swaps a day with four bays. All values are estimates from DKH-CAL-001.

Main components

Table 1. Main components. Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

BOM	Component	Concept choice	Notes
1	Cabinet body	1.5 mm galvanized steel panels with folded flanges, joined with blind rivets (laser-cut front panel, back, two sides, roof, floor pan), 1,000 x 500 x 1,400 mm on a 100 mm plinth; bay deck at 0.70 m and charger shelf below; roof frame of two 40 x 40 x 4 mm angle beams under the canopy posts	IP54 target; light powder coat; blank plates over unfitted bays
2	Bay doors (4; 2 in the first prototype)	2 mm steel doors on stainless piano hinges over 150 x 510 mm openings, with EPDM gaskets, 12 V fail-secure solenoid locks and door switches	Locked unless the controller releases them
3	Bay compartments (4; 2 in the first prototype)	Steel liner 180 x 358 x 530 mm per bay with a vent port to the plenum; cradle printed with its SwapCell guide faces, folded catch bracket with class D latch catch, and receptacle with 10 kΩ INTERLOCK coding resistor; NTC at the receptacle	Mirrors the SwapCell wall dock; pack stands connector down, lid face toward the rider
4	SwapCell packs (reference)	Interface v0.3: 46.8 V nominal, about 468 Wh, 2.85 kg, 340 x 90 x 80 mm body, 393 mm overall	Three resident with four bays, one with two; priced in the SwapCell project
5	Chargers (4; 2 in the first prototype)	Certified 54.6 V 5 A CC-CV lithium-ion chargers, about 303 W input each, power factor 0.9 or better	No custom mains electronics
6	Dock controller	ESP32, four SPI CAN controllers (one channel per bay), DC contactors, current sensors, LTE-M modem, SD card	One channel per bay because every SwapCell pack answers as node 0 by default
7	Access panel	NFC reader at 0.90 to 0.99 m, display and per-bay status lights, in a column at 0.85 to 1.21 m	Beside bay 4
8	Grid input and protection	RCBO (30 mA), a breaker per charger, surge protection and lockable isolator	Installed by a qualified electrician
9	Fire detection and suppression	Heat and smoke detector under the roof, condensed-aerosol unit strapped to the back panel at the plenum top	Aerosol can suppress flame but cannot stop a cell in runaway; vermiculite floors are tested against it (line 19)
10	Vent plenum and roof vent hood	Rear plenum 137 mm deep behind the bays, two 120 mm fans under the roof blowing through it into a vent hood with slots in its back face and a rain lip; filtered intake in the service door	Takes charger heat and any vented gas up and out at the back
11	Solar panel	400 W monocrystalline, tilted 10 degrees, front top edge at 2.45 m, rear edge underside at 2.21 m	Shades the rider and the front of the cabinet
12	Canopy frame	Four 50 x 50 x 3 mm steel posts on welded base and cap plates, bolted through the roof into the roof beams; two 50 x 40 x 3 mm rails; four panel end clamps	Front posts see about 1.25 kN of uplift at a 30 m/s gust
13	MPPT controller	Programmable lithium profile at 54.6 V, output limited to 5 A, feeding bay 1	Bay 1 keeps its AC charger as a fallback
14	Plinth and anchor frame		Anchors and pad are lines 17 and 18

BOM	Component	Concept choice	Notes
		Welded 100 x 50 x 5 mm steel channel frame flush with the body (1,000 x 500 mm), floor bolted to it, with four M12 anchor holes and socket holes above them	
15	Service door	920 x 520 mm steel door on a piano hinge over the technical compartment, with a cam lock and a filtered intake slot	Technician access only
17	Anchors (site)	Four M12 x 160 mm stainless wedge anchors, 100 mm embedment, tamper-resistant nuts	3.8 kN design tension each; not in the exploded view
18	Site pad	Cast reinforced concrete pad 1,400 x 1,100 x 300 mm, about 1,109 kg, top flush with the ground	For the first pilot on private ground; not in the exploded view
19	Vermiculite bay floor (trial option)	Stainless tray 175 x 168 x 20 mm of exfoliated vermiculite on each bay floor in front of the cradle	Propagation trial only; not fitted to the first prototype or counted in its cost

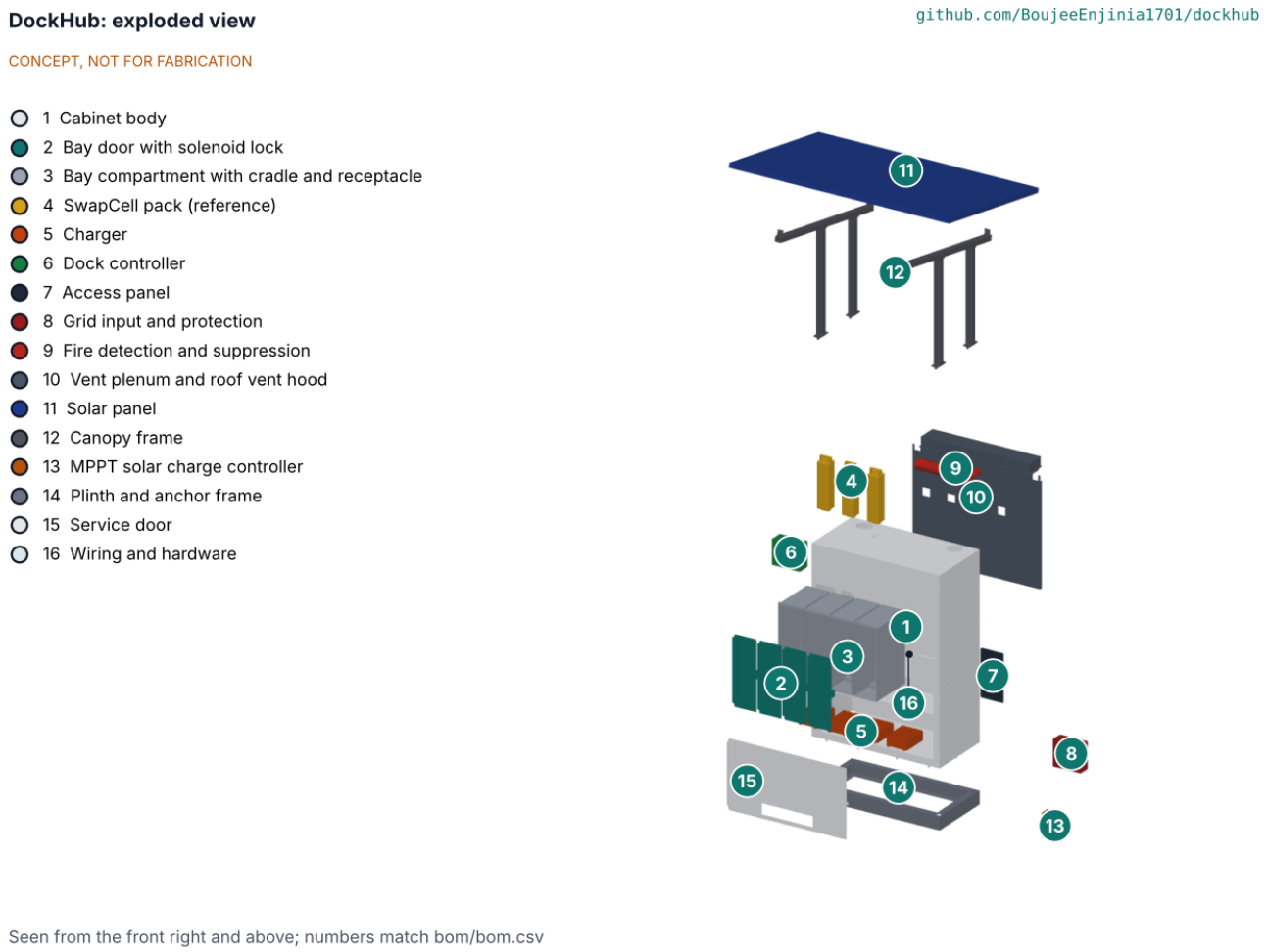


Figure 3. Exploded view of the four-bay fit with BOM numbers. Lines 16 to 19 (wiring and hardware, site anchors and pad, trial floor trays) are not shown.



Figure 4. Section through the cabinet from the front: three packs (4) in their bay compartments (3), with bay 4 empty for the next return, above the chargers (5), controller (6), MPPT controller (13) and grid input (8); the plenum (10) is behind the bays and the vent hood above.

The parametric model `cad/src/model.py` models every component as it is made or bought (DKH-DDR-003) and checks that no two overlap and that every joint touches. It exports STEP and STL files to `cad/step/` and `cad/stl/`, and the general arrangement drawing DKH-DWG-001 (Rev P2, `cad/drawings/`) shows the first-prototype fit with bays 1 and 2 fitted and bays 3 and 4 blanked.

Numbers

All values are estimates from DKH-CAL-001, printed by `docs/04-calcs/sizing.py`.

Table 2. Key numbers.

Quantity	Value	Requirement
Swap time	31 s by task analysis	R1 met
Charge time per bay	1.2 h (20 to 80 %), 2.3 h full; 1.3 h from a 15 % return to the 80 % release level	R3 met
Energy per swap	398 Wh stored, 465 Wh from the grid, about $0.093 at $0.20/kWh	
Throughput ceiling	36 swaps a day with four bays (three packs, released full); 12 with two bays	
Design day, 20 swaps	Four bays: 20 served, longest wait 4 min. Two bays: 9 served	R4 met by four bays
Daily energy at 20 swaps	8.76 kWh from the grid (0.72 kWh cabinet loads), 0.77 kWh from the panel, 7.58 kWh stored in packs	
Solar share	9.3 % of charging energy; 11.4 % if flat packs are routed to bay 1 in daylight, at the cost of one rider in 20; 16.1 % if the solar output is switched to the flattest pack (TRL 4 trial, not adopted)	R6 met (thin margin) against about 9 %
Peak grid draw	1.25 kW with four chargers; 11.0 A at 120 V and 5.7 A at 230 V with power factor 0.95	R5 met with power-factor-corrected chargers
Cabinet air at 45 °C ambient	about 49 °C (4.2 to 4.8 K rise with fans and sun on two walls)	
Charging ambient limit	Uninterrupted 5 A charging only below about 29 to 39 °C ambient; cold-soaked packs cannot start below about -1 °C	**R9 at risk** against full-rate charging up to about 30 °C
Fire response	Electronic detection latency 3.1 s; contactor opens in about 0.15 s	R7 not verifiable at TRL 3
Mass	217 kg for the two-bay prototype without packs; 247 kg for four bays with three packs; site pad about 1,109 kg	
Wind at a 30 m/s gust	1,590 N uplift and 1.5 kN drag; 2.91 kN·m overturning against 0.53 kN·m self-weight; with the pad, factor 1.12 against tipping after safety factors	R15 met with four M12 anchors and the cast pad
Parts cost	$1,519 with two of four bays fitted, anchors and pad included ($1,214 without them); $1,743 with four	**R16 over the value-engineering target by $319**

Key design choices

Items 1 to 8 are the TRL 2 recommendations, decided by Amish on 2026-09-25: go with recommendation (DKH-DDR-001, DKH-DDR-002).

1. **One SwapCell dock per bay, in a shared cabinet.** Each bay reuses the SwapCell dock cradle, charger and handshake, so DockHub adds only the cabinet, access, safety and energy supply. One certified charger and one CAN channel per bay, rather than a shared charger matrix.
2. **Four bays in one row at hand height.** Keeps the footprint to 1.0 x 0.5 m and every handle within reach (0.73 to 1.10 m). The first prototype fits bays 1 and 2 to stay close to the value-engineering target; the cabinet is built for four.

3. **Separate steel compartment per bay with a rear plenum.** A venting pack's gas and flame go to the back and up through the roof fans and vent hood, not out of the door the user is standing at, and each pack is separated from its neighbors by two steel walls and an air gap.
4. **Doors stay locked during a fire alarm.** Keeps people from opening a bay with a pack in runaway; the fire service opens the cabinet with a key. The trade-off is that a rider cannot retrieve a returned pack during an alarm; a warning light and a sign explain why.
5. **Grid first, solar as a supplement.** The 400 W canopy stays as shade and supplement, and R6 is relaxed to 10 %. DKH-CAL-001 finds 9.3 %; on 2026-10-02 Amish restated R6 as about 9 % with normal routing, and switching the solar charger to the bay with the flattest pack is a TRL 4 controller change to try (DKH-DEC-001).
6. **Solar feeds one bay directly through an MPPT controller.** Avoids an inverter and keeps all mains parts certified. It is also why the solar share is low: bay 1 can use the sun only while it holds a pack that needs charge.
7. **Offline-first access with NFC tokens.** The station keeps a token list and a log and syncs over LTE-M when it can, with an optional phone app. Each 20 B token entry records a hash of the card identifier, whether it is a named token or an anonymous prepaid card, whether a pack deposit is held, the prepaid swaps left and an expiry date (anonymous cards decided by Amish, 2026-10-02). The cabinet is unchanged.
8. **Pack health gate.** Packs below 70 % state of health, with a fault record or outside the charge temperature window are held for a technician, using the SwapCell in-pack log.

Engineering proposals made at TRL 3, decided by Amish on 2026-09-25: go with recommendation (DKH-DDR-001 items 12 and 13, DKH-DDR-002): build to SwapCell interface v0.3; release packs at 80 % state of charge or more; specify chargers with a power factor of 0.9 or better; a plinth flush with the body.

Relationship to other lab projects

- **SwapCell** (pack, interface v0.3, wall dock). DockHub builds to interface v0.3 and changes nothing in it: coded INTERLOCK receptacle (item W), dock host type 1, class D catch (item V). A multi-bay dock still needs a CAN channel per bay, because packs default to node 0 and a node number would have to change at every station. SwapCell's 45 °C charge limit is what sets DockHub's hot-weather shortfall (R9); any question about it goes to that project, not changed here.
- **TwinKit** could later collect DockHub's logs alongside other lab devices. That is a suggestion only and is not part of this concept.

Safety

> **Safety:** DockHub stores and charges lithium-ion packs of about 468 Wh each (three in the four-bay station, briefly four during a swap) next to a public walkway, from mains power. A cell in thermal runaway vents flammable, toxic gas and can ignite its neighbors; a pack fire is hard to extinguish and can re-ignite hours later. Treat the cabinet as a fire hazard at every stage of building and testing.

- **Lithium cells.** Charge only packs that pass the CAN check; never force-charge a pack that does not answer. Hold damaged, wet or faulted packs in their bay, uncharged, until a technician removes them to a fireproof container. Build and test the first cabinet on a non-combustible surface, outdoors or in a fire-rated space, with a lithium-rated extinguisher and a sand bucket at hand, and never leave it charging unattended until the detection and shutdown chain has been tested.

- **Fire containment is unproven.** The per-bay compartments and plenum are a concept. Whether they stop one pack's runaway from spreading (R8) can only be shown by a propagation test, which is TRL 4 or later work and on hold. Condensed-aerosol suppression can knock down flame but does not cool cells in runaway.
- **Mains voltage.** The technical compartment holds 120 V or 230 V AC wiring. It must be installed or checked by a qualified electrician under local electrical code, with an RCD (GFCI), per-charger breakers, surge protection, a lockable isolator and bonding of all metal parts to protective earth. Keep mains and the 54.6 V DC side physically separated and labeled. Use chargers with a power factor of 0.9 or better, or the circuit can be overloaded (DKH-CAL-001 section 6).
- **DC contacts and faults.** Receptacle contacts are at up to 54.6 V DC and a pack can deliver about 500 A into a short. Contacts stay dead until the SwapCell coded interlock and a valid heartbeat enable the pack. Each bay's DC line needs a fuse rated for 60 V DC with at least 1 kA breaking capacity.
- **Heat.** A sunlit cabinet in hot weather takes packs above their 45 °C charge limit; the pack and controller must pause charging, and the station must not be relied on for charging in the hottest hours.
- **Public siting and wind.** Place the cabinet away from building entrances, exits and windows, with the vent hood slots venting away from people. The canopy lifts and tips the cabinet in a strong gust: anchor the plinth to the cast site pad (1,400 x 1,100 x 300 mm) with the four M12 wedge anchors, each rated for at least 3.8 kN design tension, and bolt the canopy posts through to the roof frame. Keep a clear pedestrian path beside the cabinet.
- **Pinch, lifting and sharp edges.** Fold or hem all sheet edges; door locks must not pinch fingers. Canopy edges above head height need rounded corners. The cabinet weighs about 217 kg empty: move it with lifting equipment, not by hand.

Open questions

- Pack ownership model: a station-owned pool of packs exchanged one for one, as R10 assumes; to be confirmed in co-design. Decided by Amish, 2026-10-02 (DKH-DEC-001).
- First pilot site and partner: a delivery-worker group in a high-income city, New York City first, with Los Deliveristas Unidos (Workers Justice Project) as the first candidate partner to approach. Decided by Amish, 2026-10-02 (DKH-DEC-001).
- R6: restated as about 9 % with normal routing; flattest-pack solar switching to be tried at TRL 4 as a controller change. Decided by Amish, 2026-10-02 (DKH-DEC-001).
- R9: keep the pack's 45 °C charge limit and accept daytime pauses; R9 restated as full-rate charging up to about 30 °C ambient, with the cold-soak limit near -1 °C stated; active cooling only for a hot-climate site. Decided by Amish, 2026-10-02 (DKH-DEC-001).
- The two-bay first prototype holds only one pack and cannot run the duty case. Is that acceptable for a first build, given R16?
- Propagation test: a later version aims at a UL 9540A unit-level propagation test, with a heater-triggered runaway in one bay at an outdoor test site first, before paying an accredited lab. Decided by Amish, 2026-10-02 (DKH-DEC-001); whether a workshop-built cabinet can pass it is what the self-run trial will show.
- Fire suppression: keep the condensed-aerosol unit for the first prototype and test vermiculite-filled bay floors as a variant in the propagation trial; choose on the result. Decided by Amish, 2026-10-02 (DKH-DEC-001).
- Siting: the first pilot is on private property with an existing single-phase circuit (the partner's premises or a willing business frontage), not on a public sidewalk; a city sidewalk permit is sought only for a second site. Decided by Amish, 2026-10-02 (DKH-DEC-001).

- Riders who do not want to share identity: anonymous prepaid NFC cards sold by the partner, each carrying a pack deposit, alongside named tokens. Decided by Amish, 2026-10-02 (DKH-DEC-001).